

KITABU QUEST

S6

Subsidiary Mathematics

Mascot: mountain gorilla

Scene: students solving applied mathematics problems with graphs, models, and a mountain gorilla mascot
CBC aligned draft learner book

Think • Explore • Achieve

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Title Page

KITABU QUEST S6 Subsidiary Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S6 learners study Subsidiary Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Learn through examples, activities, practice, and reflection.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. COMPLEX NUMBERS
2. LOGARITHMIC AND EXPONENTIAL FUNCTIONS
3. INTEGRATION
4. ORDINARY DIFFERENTIAL EQUATIONS

As you finish each topic, return to the success criteria and correct one answer before moving on.

COMPLEX NUMBERS: Lesson Opener

Learning focus: COMPLEX NUMBERS

Country link: a Rwanda school, market, farming, building, transport, or data problem for COMPLEX NUMBERS.

By the end of this topic, you should be able to:

- perform operations on complex numbers in different forms and use them to solve related problems in Physics (voltage and current in alternating current), Computer Science(fractals), Trigonometry (Euler's formula to transform trigonometric expressions), ...
- definition and operations

Inquiry questions:

- How can complex numbers help you solve a real problem in Subsidiary Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

COMPLEX NUMBERS: Learn

COMPLEX NUMBERS teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- COMPLEX NUMBERS
- COMPLEX
- NUMBERS
- method
- model
- unit
- answer
- check

COMPLEX NUMBERS: Worked Example

Worked example for COMPLEX NUMBERS: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

COMPLEX NUMBERS: Vocabulary And Study Skill

Use these words and study moves before you practise COMPLEX NUMBERS. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- COMPLEX NUMBERS
- COMPLEX
- NUMBERS
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about COMPLEX NUMBERS without a clear example, method, or evidence.

COMPLEX NUMBERS: Guided Activity

Guided activity for COMPLEX NUMBERS:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

COMPLEX NUMBERS: Practice And Correction

Practice for COMPLEX NUMBERS:

Main outcome: perform operations on complex numbers in different forms and use them to solve related problems in Physics (voltage and current in alternating current), Computer Science(fractals), Trigonometry (Euler's formula to transform trigonometric expressions),

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to complex numbers and solve it.

COMPLEX NUMBERS: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from COMPLEX NUMBERS.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

COMPLEX NUMBERS: Outcome Check

Outcome check for COMPLEX NUMBERS: Perform operations on complex numbers in different forms and use them to solve related problems in Physics (voltage and current in alternating current), Computer Science(fractals), Trigonometry (Euler's formula to transform trigonometric expressions),

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

COMPLEX NUMBERS: Outcome Check

Outcome check for COMPLEX NUMBERS: Definition and operations.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Lesson Opener

Learning focus: LOGARITHMIC AND EXPONENTIAL FUNCTIONS

Country link: a Rwanda school, market, farming, building, transport, or data problem for LOGARITHMIC AND EXPONENTIAL FUNCTIONS.

By the end of this topic, you should be able to:

- extend the concepts of functions to investigate fully logarithmic and exponential functions and use them to model and solve problems about interest rates, population growth or decay, magnitude of earthquake, etc

Inquiry questions:

- How can logarithmic and exponential functions help you solve a real problem in Subsidiary Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Learn

LOGARITHMIC AND EXPONENTIAL FUNCTIONS teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- LOGARITHMIC AND EXPONENTIAL FUNCTIONS
- LOGARITHMIC
- EXPONENTIAL
- FUNCTIONS
- method
- model

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Worked Example

Worked example for LOGARITHMIC AND EXPONENTIAL FUNCTIONS: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Vocabulary And Study Skill

Use these words and study moves before you practise LOGARITHMIC AND EXPONENTIAL FUNCTIONS. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- LOGARITHMIC AND EXPONENTIAL FUNCTIONS
- LOGARITHMIC
- EXPONENTIAL
- FUNCTIONS
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about LOGARITHMIC AND EXPONENTIAL FUNCTIONS without a clear example, method, or evidence.

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Guided Activity

Guided activity for LOGARITHMIC AND EXPONENTIAL FUNCTIONS:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Practice And Correction

Practice for LOGARITHMIC AND EXPONENTIAL FUNCTIONS:

Main outcome: extend the concepts of functions to investigate fully logarithmic and exponential functions and use them to model and solve problems about interest rates, population growth or decay, magnitude of earthquake, etc.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to logarithmic and exponential functions and solve it.

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from LOGARITHMIC AND EXPONENTIAL FUNCTIONS.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Outcome Check

Outcome check for LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Extend the concepts of functions to investigate fully logarithmic and exponential functions and use them to model and solve problems about interest rates, population growth or decay, magnitude of earthquake, etc.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

INTEGRATION: Lesson Opener

Learning focus: INTEGRATION

Country link: a Rwanda school, market, farming, building, transport, or data problem for INTEGRATION.

By the end of this topic, you should be able to:

- : Use integration as the inverse of differentiation and as the limit of a sum and apply them to find area of a plane shapes
- basic Integration Formulas
- integration by change of variables
- integration by Parts □ Definite integrals □ Definition □ Properties □ Techniques of integration □ Applications of definite integrals

Inquiry questions:

- How can integration help you solve a real problem in Subsidiary Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

INTEGRATION: Learn

INTEGRATION teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- INTEGRATION
- method
- model
- unit
- answer
- check

INTEGRATION: Worked Example

Worked example for INTEGRATION: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

INTEGRATION: Vocabulary And Study Skill

Use these words and study moves before you practise INTEGRATION. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- INTEGRATION
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about INTEGRATION without a clear example, method, or evidence.

INTEGRATION: Guided Activity

Guided activity for INTEGRATION:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

INTEGRATION: Practice And Correction

Practice for INTEGRATION:

Main outcome: : Use integration as the inverse of differentiation and as the limit of a sum and apply them to find area of a plane shapes.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to integration and solve it.

INTEGRATION: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from INTEGRATION.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

INTEGRATION: Outcome Check

Outcome check for INTEGRATION: : Use integration as the inverse of differentiation and as the limit of a sum and apply them to find area of a plane shapes.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

INTEGRATION: Outcome Check

Outcome check for INTEGRATION: Basic Integration Formulas.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

ORDINARY DIFFERENTIAL EQUATIONS: Lesson Opener

Learning focus: ORDINARY DIFFERENTIAL EQUATIONS

Country link: a Rwanda school, market, farming, building, transport, or data problem for ORDINARY DIFFERENTIAL EQUATIONS.

By the end of this topic, you should be able to:

- use ordinary differential equations of first to model and solve related problems in Physics, Economics, Chemistry, Biology, ...
- the right hand side is equal to zero □ Applications:

Inquiry questions:

- How can ordinary differential equations help you solve a real problem in Subsidiary Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

ORDINARY DIFFERENTIAL EQUATIONS: Learn

ORDINARY DIFFERENTIAL EQUATIONS teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- ORDINARY DIFFERENTIAL EQUATIONS
- ORDINARY
- DIFFERENTIAL
- EQUATIONS
- method
- model
- unit
- answer

ORDINARY DIFFERENTIAL EQUATIONS: Worked Example

Worked example for ORDINARY DIFFERENTIAL EQUATIONS: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

ORDINARY DIFFERENTIAL EQUATIONS: Vocabulary And Study Skill

Use these words and study moves before you practise ORDINARY DIFFERENTIAL EQUATIONS. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- ORDINARY DIFFERENTIAL EQUATIONS
- ORDINARY
- DIFFERENTIAL
- EQUATIONS
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about ORDINARY DIFFERENTIAL EQUATIONS without a clear example, method, or evidence.

ORDINARY DIFFERENTIAL EQUATIONS: Guided Activity

Guided activity for ORDINARY DIFFERENTIAL EQUATIONS:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

ORDINARY DIFFERENTIAL EQUATIONS: Practice And Correction

Practice for ORDINARY DIFFERENTIAL EQUATIONS:

Main outcome: use ordinary differential equations of first to model and solve related problems in Physics, Economics, Chemistry, Biology,

1. Solve one ordinary differential equations question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to ordinary differential equations and solve it.

ORDINARY DIFFERENTIAL EQUATIONS: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from ORDINARY DIFFERENTIAL EQUATIONS.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

ORDINARY DIFFERENTIAL EQUATIONS: Outcome Check

Outcome check for ORDINARY DIFFERENTIAL EQUATIONS: Use ordinary differential equations of first to model and solve related problems in Physics, Economics, Chemistry, Biology,

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

ORDINARY DIFFERENTIAL EQUATIONS: Outcome Check

Outcome check for ORDINARY DIFFERENTIAL EQUATIONS: The right hand side is equal to zero

□ Applications:

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

COMPLEX NUMBERS: Review Clinic 1

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from COMPLEX NUMBERS that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Review Clinic 2

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from LOGARITHMIC AND EXPONENTIAL FUNCTIONS that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

INTEGRATION: Review Clinic 3

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from INTEGRATION that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

ORDINARY DIFFERENTIAL EQUATIONS: Review Clinic 4

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from ORDINARY DIFFERENTIAL EQUATIONS that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Subsidiary Mathematics.

- COMPLEX NUMBERS: Explain this term using an example from the book, your school, or your community.
- LOGARITHMIC AND EXPONENTIAL FUNCTIONS: Explain this term using an example from the book, your school, or your community.
- INTEGRATION: Explain this term using an example from the book, your school, or your community.
- ORDINARY DIFFERENTIAL EQUATIONS: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Subsidiary Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Subsidiary Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.